

- 2 (a) sum of kinetic and potential energies of molecules / particles / atomsM1
 random (distribution) A1 [2]
- (b) $+\Delta U$: increase in internal energy B1
 $+q$: heating of / heat supplied to system B1
 $+w$: work done on system B1 [3]
- (c) (i) work done = $p\Delta V$ C1
 $= 1.0 \times 10^5 \times (2.1 - 1.8) \times 10^{-3}$
 $= 30 \text{ J}$ M1
 $w = 30 \text{ J}, q = 0$ so $\Delta U = 30 \text{ J}$ A1 [3]
- (ii) these three marks were removed, as insufficient data was given in the question.

[Total: 8]

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2 (a) straight line through origin